

Model-A Propagation-Aware Bragg Transfer Function

Square and Gaussian pulses at $L = 0, 150$ m and 2000 m

MIGA programme

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Abstract

This manual documents the area-compensated propagation model implemented in the propagation-aware Python calculator. In a single-frequency retroreflected geometry, the forward optical envelope and its reflected copy reach the atoms with the round-trip separation $\tau_d = 2L/c$. Their overlap determines the effective two-photon Rabi envelope. Model A rescales every overlapped pulse to restore the nominal $\pi/2$ – π – $\pi/2$ areas, representing an experiment in which pulse areas are recalibrated after propagation is included. The manual derives the result for square and Gaussian optical-intensity pulses, explains the sensitivity and transfer functions, and compares reproducible first-order examples at $L = 0, 150$ m and 2000 m. All three use $T = 250$ ms, $\tau_\pi = 50$ μ s and $\tau_{\pi/2} = 25$ μ s. For the Gaussian case, these two widths are intensity FWHM values.

i Main result

Area compensation affects square and Gaussian pulses differently. A delayed square pulse becomes shorter and must become higher, which changes the high-frequency envelope of H_{AI} . An untruncated Gaussian remains a Gaussian of the same width; Model-A normalization exactly cancels its overlap attenuation, leaving only a shift by $\tau_d/2$. Consequently, the magnitude $|H_{AI}|$ is essentially unchanged for the Gaussian case, apart from negligible effects of numerical tail truncation.

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1 Purpose and model boundary

The calculator describes a resonant three-pulse Mach–Zehnder sequence with nominal areas $\pi/2-\pi-\pi/2$. It extends the finite-pulse sensitivity-function treatment of Cheinet *et al.*[1], the MIGA phase-noise analysis[2], and the pulse-shaping description based on accumulated Rabi area[3]. This manual considers only the area-compensated setting labelled A: area compensated in the interface.

The backend is an effective resonant two-level model. The Bragg order n multiplies the phase response but does not alter the normalized pulse dynamics. The calculation includes delayed envelope overlap, pulse-area recalibration, finite-pulse filtering and laser phase/frequency-noise conversion. It does not include the momentum-state ladder, velocity selectivity, diffraction phase, AC Stark shifts, Doppler detuning, spontaneous emission, parasitic interferometers or cavity dynamics. These effects require a multi-state Bragg treatment such as Refs. [4, 5].

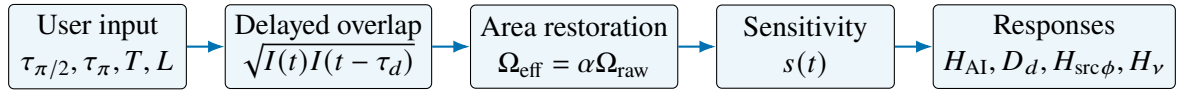


Figure 1. Model-A calculation chain. The delayed optical overlap determines a raw Rabi envelope. Its area is restored before the sensitivity and transfer functions are evaluated.

2 Quick start

Launch the interface from the project directory with

```
python3 bragg_transfer_function_propagation_ui.py
```

Select A: area compensated in the propagation-model menu. Choose Square or Gaussian, enter the $\pi/2$ and π widths in microseconds, and enter the centre-to-centre pulse separation T and one-way atom-to-retroreflector distance L . A square width is a command duration. A Gaussian width is the FWHM of the optical-intensity envelope. Run the built-in validation suite with

```
python3 bragg_transfer_function_propagation_ui.py --self-test
```

3 Geometry and delayed two-photon coupling

Let $I(t)$ and $\phi(t)$ denote the source-side intensity envelope and optical phase. Taking the forward field at the atoms as the time reference, the reflected field samples the source a round-trip time earlier,

$$\tau_d = \frac{2L}{c}. \quad (1)$$

After removing constant propagation phases and assuming identical spatial modes, the complex two-photon coupling is

$$\tilde{\Omega}(t) = \Omega_{\text{in}} \frac{\sqrt{I(t)I(t-\tau_d)}}{I_{\text{pk}}} \exp\{i[\phi(t) - \phi(t-\tau_d)]\}. \quad (2)$$

Its modulus gives the raw envelope

$$\Omega_{\text{raw}}(t) = \Omega_{\text{in}} \frac{\sqrt{I(t)I(t-\tau_d)}}{I_{\text{pk}}}, \quad (3)$$

while its argument gives the local counter-propagating phase. Envelope overlap modifies the pulse shape and therefore H_{AI} . Phase differencing produces the separate delay filter D_d derived in section 7. The two effects enter at different stages of the calculation.

Definition of L

The software uses $\tau_d = 2L/c$, where L is one way. Entering a path length that already includes the round trip counts the factor of two twice. Additional fibre, electronic or cavity delays are not included automatically.

4 Square and Gaussian overlap

4.1 Square pulse

For a unit-height square command of duration τ centred on t_c , the common interval has duration

$$\tau_{\text{ov}} = \tau - \tau_d, \quad \tau > \tau_d, \quad (4)$$

and is centred at $t_c + \tau_d/2$. The raw rate is Ω_{in} inside that interval, so the area ratio is

$$\frac{A_{\text{raw}}}{A_0} = 1 - \frac{\tau_d}{\tau}. \quad (5)$$

If $\tau_d \geq \tau$, there is no common support and the software stops with an error.

4.2 Gaussian pulse

The Gaussian input convention is

$$I(t) = I_{\text{pk}} \exp\left[-\frac{(t - t_c)^2}{2\sigma^2}\right], \quad \tau_{\text{FWHM}} = 2\sqrt{2 \ln 2} \sigma. \quad (6)$$

Substitution into [equation \(3\)](#) gives

$$\Omega_{\text{raw}}(t) = \Omega_{\text{in}} \exp\left[-\frac{\tau_d^2}{8\sigma^2}\right] \exp\left[-\frac{(t - t_c - \tau_d/2)^2}{2\sigma^2}\right]. \quad (7)$$

The overlap is therefore a Gaussian of the same width, shifted by $\tau_d/2$. Its peak attenuation is the first exponential in [equation \(7\)](#). The calculation truncates each input envelope at $\pm 6\sigma$. The common numerical support then has total duration $12\sigma - \tau_d$, and its small missing-tail correction is retained through the error function.

5 Model A: pulse-area restoration

For every interaction j , Model A defines

$$\alpha_j = \frac{A_j^{\text{target}}}{A_j^{\text{raw}}}, \quad \Omega_{\text{eff}j}(t) = \alpha_j \Omega_{\text{raw}j}(t), \quad A_j^{\text{target}} \in \{\pi/2, \pi, \pi/2\}. \quad (8)$$

It represents an experimental recalibration of the local pulse area after propagation has been included.

For a square pulse, [equation \(5\)](#) gives

$$\alpha_{\text{sq}} = \frac{\tau}{\tau - \tau_d}, \quad \Omega_{\text{eff,pk}} = \Omega_{\text{in}} \alpha_{\text{sq}}. \quad (9)$$

The compensated pulse is shorter and higher than the no-delay pulse, so its Fourier envelope changes.

For an untruncated Gaussian, the area attenuation is exactly the peak attenuation. Hence

$$\alpha_G = \exp\left[\frac{\tau_d^2}{8\sigma^2}\right], \quad \Omega_{\text{eff}G}(t) = \Omega_{\text{in}} \exp\left[-\frac{(t - t_c - \tau_d/2)^2}{2\sigma^2}\right]. \quad (10)$$

Thus Model A restores the original Gaussian exactly, except for the common time shift. In the implemented $\pm 6\sigma$ representation, the normalization also includes the ratio of the nominal and common-support error functions. Because the removed tails are extremely small in the present example, the numerical result is indistinguishable from [equation \(10\)](#) at the plotted precision.

6 Sensitivity function

Define the accumulated area of interaction j as

$$\Theta_j(t) = \int_{t_{j,s}}^t \Omega_{\text{eff}j}(t') dt'. \quad (11)$$

Because Model A restores ideal areas, the MIGA/Cheinet sign convention gives

$$s(t) = \begin{cases} \sin \Theta_1(t), & t \text{ in the first } \pi/2 \text{ pulse,} \\ +1, & \text{between pulses 1 and 2,} \\ \cos \Theta_2(t), & t \text{ in the } \pi \text{ pulse,} \\ -1, & \text{between pulses 2 and 3,} \\ -\cos \Theta_3(t), & t \text{ in the last } \pi/2 \text{ pulse,} \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

The program evaluates the equivalent resonant two-level phase-step response. The two constructions agree for the restored $\pi/2 - \pi - \pi/2$ areas.

7 Transfer functions

Using

$$\mathcal{F}[x](\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt, \quad \omega = 2\pi f, \quad (13)$$

the atomic response to local Bragg phase is

$$H_{\text{AI}}(f) = n \int_{-\infty}^{\infty} \dot{s}(t) e^{-i2\pi f t} dt = n i 2\pi f \int_{-\infty}^{\infty} s(t) e^{-i2\pi f t} dt. \quad (14)$$

The second form follows by integration by parts because $s(t)$ vanishes outside the sequence. The code uses this second, numerically stable form. The delay between the two source-phase samples gives

$$D_d(f) = 1 - e^{-i2\pi f \tau_d}, \quad |D_d| = 2|\sin(\pi f \tau_d)|. \quad (15)$$

The source-phase and source-frequency responses are

$$H_{\text{src}\phi}(f) = H_{\text{AI}}(f) D_d(f), \quad H_{\nu}(f) = \frac{H_{\text{src}\phi}(f)}{if}. \quad (16)$$

The factor 2π cancels when frequency noise is specified in hertz because $\dot{\phi} = 2\pi\delta\nu$.

Table 1. Inputs, outputs and units of the displayed responses.

Quantity	Input → output	Unit	Interpretation
H_{AI}	local Bragg phase → atom phase	dimensionless	Atomic sequence and pulse-shape filtering
D_d	source phase → local phase difference	dimensionless	Round-trip delay differencer
$H_{\text{src}\phi}$	source phase → atom phase	dimensionless	Product $H_{\text{AI}}D_d$
H_ν	source frequency in Hz → atom phase	rad/Hz	Product $H_{\text{src}\phi}/(if)$

For a one-sided input PSD $S_x(f)$, the output PSD and variance are

$$S_\Phi(f) = |H_x(f)|^2 S_x(f), \quad \sigma_\Phi^2 = \int_0^\infty S_\Phi(f) df. \quad (17)$$

An amplitude spectral density is multiplied by $|H_x|$, not by $|H_x|^2$.

8 Numerical implementation

The schedule builder converts entered widths to nominal peak Rabi rates. For square pulses, $\Omega_{\pi/2} = (\pi/2)/\tau_{\pi/2}$ and $\Omega_\pi = \pi/\tau_\pi$. For Gaussian pulses, the target area is divided by $\sigma\sqrt{2\pi}\text{erf}(6/\sqrt{2})$. The delayed overlap, raw area and Model-A scaling are then calculated separately for each interaction.

The transfer routine evaluates $\int s(t)e^{-i\omega t} dt$. It uses Gauss–Legendre quadrature inside the pulses and integrates the long constant plateaus analytically with

$$\int_0^d e^{iqt} dt = d e^{iqd/2} \text{sinc}\left(\frac{qd}{2}\right), \quad \text{sinc}(x) = \frac{\sin x}{x}. \quad (18)$$

NumPy uses $\text{sinc}(x) = \sin(\pi x)/(\pi x)$, so the implementation passes $qd/(2\pi)$. Frequencies are processed in chunks and the quadrature order grows with the product of maximum frequency and pulse support.

✓ Validation

The built-in self-test verifies the zero-delay limit, the square overlap formula, Model-A area restoration, the analytic delay factor, $H_{\text{src}\phi} = H_{\text{AI}}D_d$, and $H_\nu = H_{\text{src}\phi}/(if)$ away from DC.

9 Worked comparison: $L = 0, 150$ m and 2000 m

9.1 Common inputs

The three examples use the same $n = 1$, $T = 250$ ms, $25\ \mu\text{s}$ $\pi/2$ width and $50\ \mu\text{s}$ π width. For square pulses, these are command durations and both nominal rates are $\Omega/(2\pi) = 10$ kHz. For Gaussian pulses, they are intensity FWHM values, giving $\sigma_{\pi/2} = 10.616\ 523\ \mu\text{s}$, $\sigma_\pi = 21.233\ 045\ \mu\text{s}$ and the common area-normalized input peak $\Omega_0/(2\pi) = 9.3944$ kHz. The frequency range is 0.1 Hz–100 kHz.

The round-trip delays are

$$\tau_d(L = 0) = 0, \quad \tau_d(150\text{ m}) = 1.000\ 692\ \mu\text{s}, \quad \tau_d(2000\text{ m}) = 13.342\ 564\ \mu\text{s}. \quad (19)$$

$L = 0$ is therefore the no-propagation reference, whereas 2000 m is a deliberately demanding case in which the delay is no longer small relative to the $25\ \mu\text{s}$ pulse.

Table 2. Model-A square-pulse overlap and area compensation. The effective peak is the nominal 10 kHz rate multiplied by α .

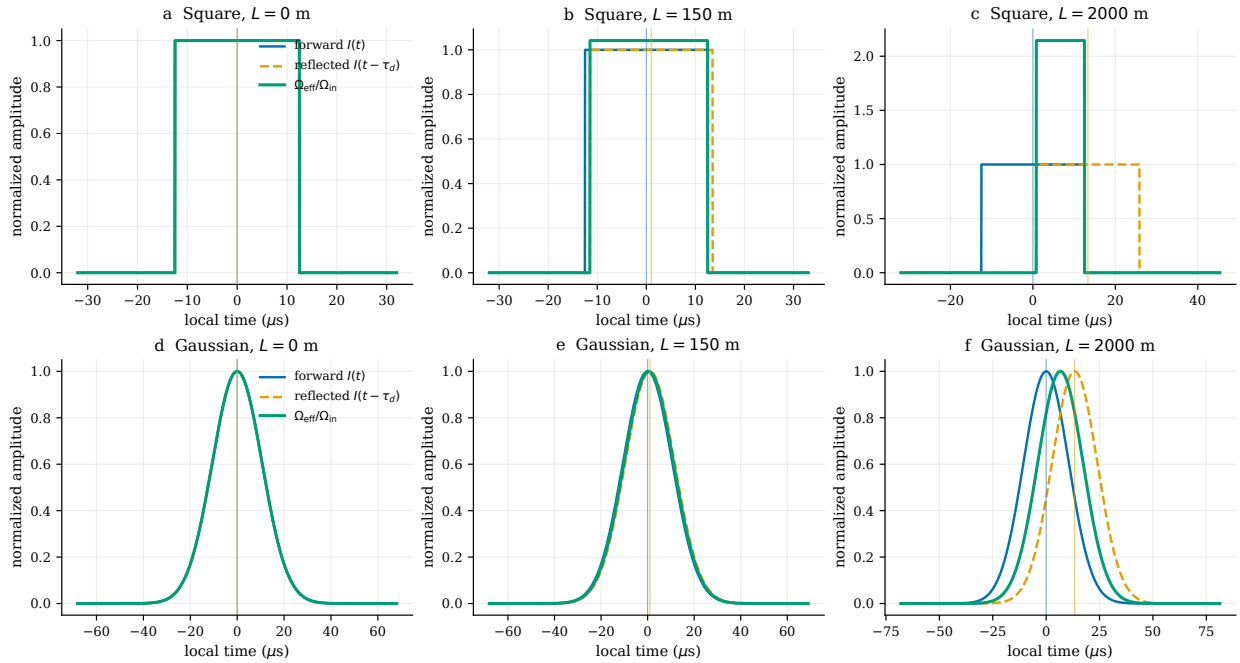
L (m)	$\tau_{\pi/2}^{\text{ov}}$ (μs)	$\alpha_{\pi/2}$	Peak $\pi/2$ (kHz)	τ_{π}^{ov} (μs)	α_{π}	Peak π (kHz)
0	25.000000	1.000000	10.0000	50.000000	1.000000	10.0000
150	23.999308	1.041697	10.4170	48.999308	1.020423	10.2042
2000	11.657436	2.144554	21.4455	36.657436	1.363980	13.6398

At 150 m, the required square compensation is mild. At 2000 m, however, only 11.657 436 μs of the square $\pi/2$ pulse overlaps, so Model A requires $\alpha_{\pi/2} = 2.144554$ and an effective rate of 21.4455 kHz. This is mathematically valid but experimentally demanding: the available optical power and validity of the effective two-level approximation must be checked.

Table 3. Gaussian overlap before and after Model-A compensation. “Support” is the retained common $\pm 6\sigma$ numerical interval, not the FWHM. The raw ratio is the uncompensated effective peak divided by the input peak.

L (m)	Raw $\pi/2$	$\alpha_{\pi/2}$	Support $\pi/2$ (μs)	Raw π	α_{π}	Support π (μs)
0	1.000000	1.000000	127.398270	1.000000	1.000000	254.796540
150	0.998890	1.001111	126.397578	0.999722	1.000278	253.795848
2000	0.820834	1.218274	114.055706	0.951840	1.050597	241.453976

For the 2000 m Gaussian $\pi/2$ pulse, the uncompensated peak falls to 0.820834 of its input value and Model A applies $\alpha = 1.218274$. After compensation, the effective Gaussian is restored to the original width and area, apart from its shift by $\tau_d/2$ and the negligible retained-tail truncation.

**Figure 2.** Pulse overlap for all six Model-A cases. The top row shows square pulses at $L = 0, 150$ m and 2000 m; the bottom row shows the corresponding Gaussian pulses. Increasing L shortens the common support. The compensated square becomes narrower and taller, while the compensated Gaussian retains its width and is shifted by $\tau_d/2$.

9.2 Atomic transfer function

At frequencies of order $1/T$, all three distances and both shapes are almost indistinguishable because the long free evolution dominates the microsecond pulse edges. At higher frequency, the envelope depends strongly on pulse shape. The square response develops sinc-like lobes and retains more high-frequency content. The smooth Gaussian response decays much more rapidly.

For the propagated square pulse, the shorter compensated duration shifts the high-frequency lobes relative to the no-delay calculation. For the Gaussian pulse, [equation \(10\)](#) implies

$$H_{\text{AI,G}}^{(L)}(f) = e^{-i2\pi f \tau_d/2} H_{\text{AI,G}}^{(0)}(f) \quad (20)$$

in the untruncated limit. The propagation changes its complex phase but not its magnitude.

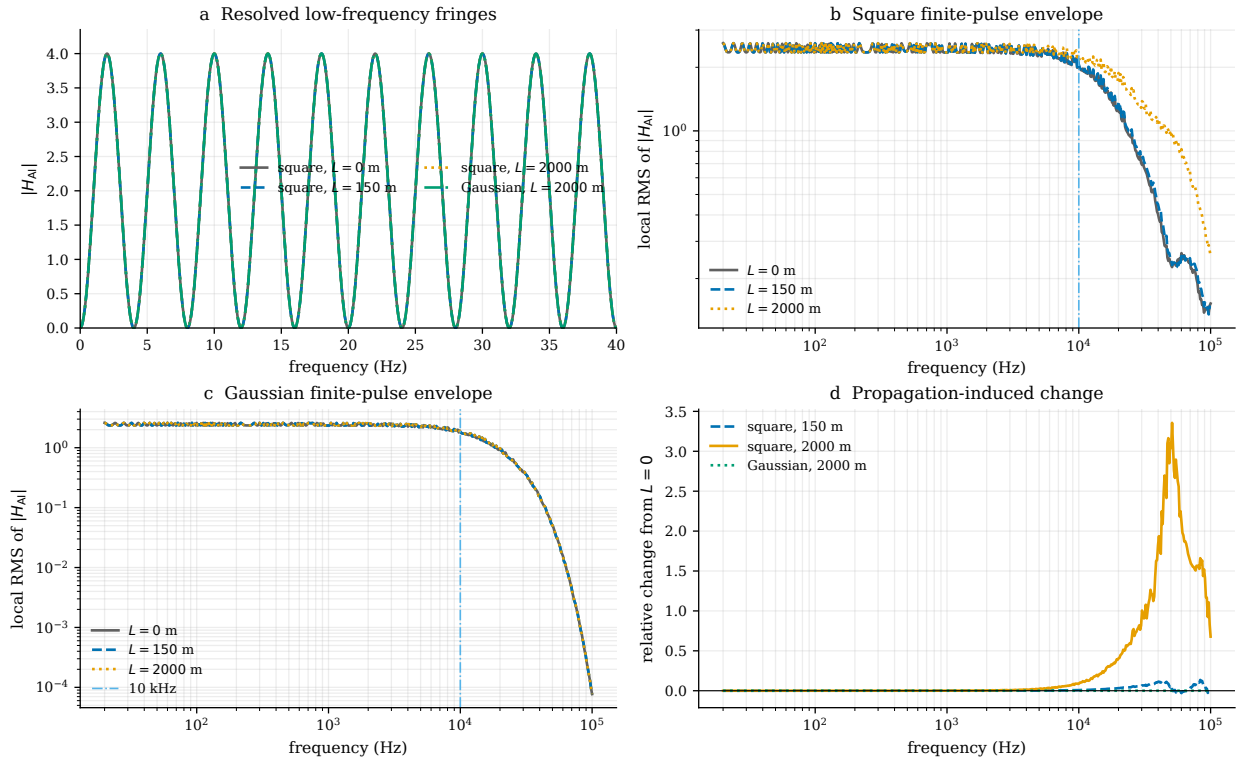


Figure 3. Atomic transfer functions under Model A. (a) The resolved low-frequency fringes coincide. (b) Square-pulse propagation changes the finite-pulse envelope, strongly for 2000 m. (c) All Gaussian magnitudes overlap because area compensation restores the Gaussian shape. (d) Relative to $L = 0$, the Gaussian magnitude is essentially invariant, whereas the square difference grows near high-frequency lobes and zeros. Local RMS is used only to display the envelope without sensitivity to isolated fringe zeros.

Table 4. Selected ± 2 Hz local-RMS values of $|H_{\text{AI}}|$. The RMS window is a plotting diagnostic, not an additional physical filter.

Shape and frequency	$L = 0$	$L = 150$ m	$L = 2000$ m
Square, 10 kHz	2.054095	2.070906	2.249210
Square, 100 kHz	$1.452\,252 \times 10^{-1}$	$1.420\,195 \times 10^{-1}$	$2.618\,094 \times 10^{-1}$
Gaussian, 10 kHz	1.846185	1.846185	1.846185
Gaussian, 100 kHz	$7.707\,363 \times 10^{-5}$	$7.707\,417 \times 10^{-5}$	$7.708\,313 \times 10^{-5}$

The square response changes progressively with distance because the compensated common interval becomes shorter. The Gaussian values remain equal to the displayed precision: under Model A, distance contributes

a time shift to the effective Gaussian and therefore a phase factor to H_{AI} , but no magnitude penalty. Its much smaller 100 kHz value is smooth pulse-shape filtering, not an extra propagation loss.

9.3 Delay filter and total response

The delay factor depends on L but not on pulse shape. Its numerical comparison is

Table 5. Magnitude and first non-DC zero of the propagation-delay differencer.

L (m)	$ D_d $ at 10 kHz	$ D_d $ at 100 kHz	First non-DC zero
0	0	0	identically zero
150	0.062865	0.618448	999.308 kHz
2000	0.814003	1.734943	74.948 kHz

For $L = 0$, $D_d = 0$ at every frequency, hence both $H_{\text{src}\phi}$ and H_v vanish; the zero response is annotated rather than drawn on logarithmic axes. The 150 m first zero remains outside the plotted band, whereas the 2000 m zero at 74.948 kHz is visible. Below the first zero, $|D_d| \simeq 2\pi f \tau_d$, so the low-frequency $|H_v|$ plateau grows linearly with L .

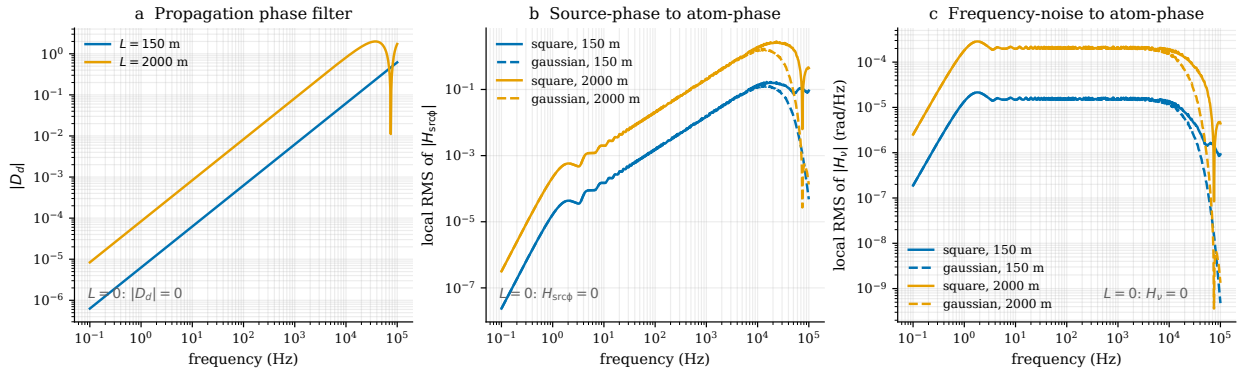


Figure 4. Combined propagation responses for 150 m and 2000 m; the $L = 0$ responses are identically zero. (a) The delay differencer depends only on L . (b) The source-phase response is $|H_{\text{src}\phi}| = |H_{\text{AI}} D_d|$; its shape dependence is inherited from the atomic filter. (c) The frequency-noise response is $|H_v| = |H_{\text{src}\phi}|/f$ in rad/Hz. Dotted levels show the ideal-pulse low-frequency local-RMS limits $2\pi\tau_d\sqrt{6}$ for $n = 1$.

i Interpretation of the three-distance comparison

$L = 0$ isolates the ordinary atomic transfer function but has no differential propagation coupling. At 150 m, square-pulse distortion and area compensation are small, while the delay factor is already nonzero. At 2000 m, square overlap is severely shortened, the required $\pi/2$ Rabi rate more than doubles, and the delay-filter zero enters the plotted band. Gaussian Model-A compensation still restores $|H_{\text{AI}}|$, but both pulse shapes retain the same unavoidable delayed sampling of source phase through D_d .

10 Reproducing the figures

From this manual directory, run

```
python3 manual_figures_propagation.py
pdflatex -interaction=nonstopmode bragg_transfer_function_propagation_manual.tex
```

```
pdflatex -interaction=nonstopmode bragg_transfer_function_propagation_manual.tex
```

The plotting script imports the same numerical functions as the GUI, writes vector PDF and preview PNG figures, and regenerates `case_values.tex`. Upload the $\text{\LaTeX}$ source, `case_values.tex`, and the `figures` folder together when compiling on Overleaf.

11 Troubleshooting and limitations

Observation	Explanation and action
No square-pulse overlap	$\tau_d \geq \tau$ for at least one pulse. Reduce L or lengthen the pulse.
Gaussian support looks much longer than its entered width	The entered width is FWHM; the numerical support retains the tails out to $\pm 6\sigma$.
Gaussian propagation curves overlap exactly	This is the expected Model-A result in equation (10) , not a missing calculation. Inspect complex phase to observe the $\tau_d/2$ shift.
Square relative-change spikes	The no-delay response is near a lobe zero. Compare complex responses or a local envelope rather than a pointwise ratio.
H_{AI} is dimensionless but H_{ν} is rad/Hz	They have different inputs: phase/phase versus phase/frequency.
Required square peak is too high	The area-restoring factor grows as $\tau/(\tau - \tau_d)$. The assumed compensation may exceed available optical power.
Result differs from a multi-order simulation	The present backend is an effective resonant two-level model and excludes unwanted momentum orders and diffraction phases.

12 File map

File	Role
<code>bragg_transfer_function_propagation_ui.py</code>	GUI, delayed pulse construction, sensitivity function and transfer functions.
<code>PROPAGATION_MODEL.md</code>	Concise model definition and launch instructions.
<code>bragg_transfer_function_propagation_manual.tex</code>	This Model-A square/Gaussian manual.
<code>manual_figures_propagation.py</code>	Deterministic generator for the case figures and numerical macros.

References

- [1] P. Cheinet *et al.*, Measurement of the sensitivity function in a time-domain atomic interferometer, *IEEE Trans. Instrum. Meas.* **57**, 1141–1148 (2008). doi:[10.1109/TIM.2007.915148](#).
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- [3] B. Fang *et al.*, Improving the phase response of an atom interferometer by means of temporal pulse shaping, *New J. Phys.* **20**, 023020 (2018). doi:[10.1088/1367-2630/aaa37c](#).
- [4] J.-N. Siemß *et al.*, Analytic theory for Bragg atom interferometry based on the adiabatic theorem, *Phys. Rev. A* **102**, 033709 (2020). doi:[10.1103/PhysRevA.102.033709](#).
- [5] J. Jenewein *et al.*, Multiport atom interferometry with Bragg pulses, *Phys. Rev. A* **105**, 063316 (2022). doi:[10.1103/PhysRevA.105.063316](#).